DR LIAM R. J. SCARRATT

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PROFILE

I am an accomplished Chemistry educator and researcher with a proven track record in curriculum development, program management, and science communication. My extensive experience includes leading and mentoring students, designing and delivering innovative chemistry courses, and pioneering interdisciplinary collaborations. I have honed my ability to convey complex scientific concepts through numerous conference presentations and publications in peer-reviewed journals. My project management skills have been demonstrated through successful science outreach initiatives and grant-funded projects. I am dedicated to fostering student success and continuously enhancing educational practices through professional development and the integration of advanced teaching technologies.

PROFESSIONAL EXPERIENCE

UNSW College

Leading Education Professional Science — September 2024 - Present
Senior Education Professional Science — January 2024 - August 2024

Course Leadership:

- Coordinated and delivered Foundation Studies and Diploma Chemistry courses, including lectures, tutorials, consultations and laboratory sessions. Managed staff assigned to tutorials and laboratory sessions and liaised with organisational contacts for successful implementation of exam writing/marking and laboratory timetabling and staffing.

Curriculum & Assessment Development:

- Redesigned the Foundation Studies Chemistry course lecture slides for improved student readability. Implemented further changes to lectures slides alongside enhancing student comprehension of laboratory instructions as one of the Chemistry team members for the FS2025 curriculum review.

Professional Development:

- Pioneered a collaboration between Chemistry and English toward the enhancement of teaching practice and curriculum design. Learned and adapted AI tools to maximise student engagement in tutorials and lectures.

Collaborative Communication:

- Trained staff in the use of key technologies, such as Gradescope, Onenote, and Microsoft Power Automate. Actively disseminates new insights to peers to promote student engagement and achievement of learning outcomes.

Institutional Engagement:

- Presented at the Academic Collaboration and Innovation (ACI) Day to share my past learnings on the communication of science. Actively engaged in ACI Breakfast Clubs, including presenting alongside the English Department as part of our active collaboration.

University of Geneva, Switzerland and University of Bristol, UK

Postdoctoral Researcher — March 2019 – August 2023

Leadership and Supervision:

- Primary supervisor for an undergraduate final year project student, overseeing project design, research skill development, and thesis production. Trained and mentored multiple Masters and PhD students in specialized experimental techniques, enhancing their research capabilities and academic progress.

Curriculum Development and Teaching:

- Developed and delivered undergraduate Chemistry laboratory classes, guiding 1st-year students through weekly lab sessions, assessing performance, and ensuring experiment viability. Contributed to curriculum validation by liaising with course designers and instructors, ensuring high-quality educational experiences.

Science Communication and Outreach:

- Communicated research in a short story format at the Futures European Researcher's Night, Bristol, UK (2021), and performed Highschool Chemistry demonstration lectures at the University of Bristol, UK (2022).
- President of the University of Geneva Postdoctoral Association, developed the format for two online research conferences during COVID.
- Part of the organizing team for SciFilmIt Hackathon Geneva (2020) and mentored at SciFilmIt Hackathon Lausanne (2021), giving an invited talk.

Project Management:

- Led a five-person team to establish the first UK SciFilmIt Hackathon, securing $28,000 through a competitive Research England grant for Enhancing Research Culture (2022-2023).

Research and Collaboration:

- Recipient of the highly competitive Marie Skłodowska-Curie Postdoctoral Fellowship (success rate of 10%) for an independent research project at the University of Bristol, UK (2021-2023). Awarded the Langmuir Prize for best speaker at the 33rd European Colloid and Interface Society Conference, Leuven, Belgium (2019).
- Published four research papers in peer-reviewed journals, refining complex experimental techniques using atomic force microscopy. Established extensive academic networks across the UK, Europe, and Australia through conference presentations, invited seminars, and collaborative research proposals.

EDUCATION

The University of Sydney

Doctor of Philosophy (Chemistry) February 2015 – January 2019

Research Focus:

- Developed a novel method to probe the slipperiness of nanofilms using atomic force microscopy.
- Created a new type of superhydrophobic surface using the wrinkling of Teflon nanofilms on shrinkable plastic sheets.

Key Achievements:

- Published four research papers in peer-reviewed journals.
- Mentored three Undergraduate Summer Scholars on independent research projects (2015, 2016, 2018).
- School of Chemistry Le Fèvre Postgraduate Student Lecturer (2018). Healy Hunter Medal for best talk at Australasian Colloid and Interface Society Student Conference (2018).
- Semi-Finalist in Famelab New South Wales, Australia (2018). 2nd Place, The University of Sydney 3 Minute Thesis Finals, NSW, Australia (2015).
- German Academic Exchange Service DAAD and Australian Nanotechnology Network scholarships for a 3-month stay at The Max Planck Institute for Polymer Research, Mainz, Germany (2017).
- Faculty of Science Postgraduate Research Prize for Outstanding Academic Achievement (2016). Australian Postgraduate Research Award (2015).

Postgraduate Teaching Fellowship & Laboratory Demonstration:

- Awarded a Teaching Fellowship with the School of Chemistry to teach first-year Chemistry tutorials (2016 – 2017).
- Supervised and demonstrated for 1st-year Special Students Program Chemistry lab classes, training students in research skills (2016-2018). Supervised and demonstrated for standard 1st-year and 3rd-year physical Chemistry lab classes, focusing on experimental techniques, data analysis, and presentation.

Bachelor of Science (Honours I) March 2010 – November 2014

- Majors: Physics and Chemistry, with first class Honours in Chemistry
- Selected to take part in the Year In Industry Program at the National Industrial Chemicals Notification and Assessment Scheme (NICNAS), working as a Regulatory Scientist (2012-2014)
- Represented the University of Sydney with music group The Acappelicans in Hong Kong (2014)
- Band Director of The University of Sydney's Science Revue (2015), and Guitarist in the Science Revue (2012, 2013)